

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 2: Case Study**

**COURSE NAME:** Cloud Computing and Big Data Analytics      **COURSE CODE:** CSA1504

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**COURSE OUTCOME COVERED**

**CO3:** *Implement cloud-based solutions using cloud storage, web APIs, and platform services for realworld applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)  
Question Count: 3

**Total Marks: 30**

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**Code of Conduct**

I **Krishna chaitanya/ 192372241** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: \_\_\_\_\_

**Assessment Tasks**

**Case Study**

**Case Study 1: Smart Retail Inventory Management System**

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations. **Smart Retail Inventory Management System**

**1. Cloud services used:**

Cloud storage, Web APIs, platform analytics services, and demand forecasting services are used.

**2. Role of Web APIs:**

APIs connect POS systems and dashboards with centralized cloud services for data exchange.

**3. Role of cloud storage:**

It securely stores product data, transaction logs, and analytics datasets.

**4. Role of platform services:**

Platform services analyze inventory data and support demand forecasting.

**5. Security mechanisms:**

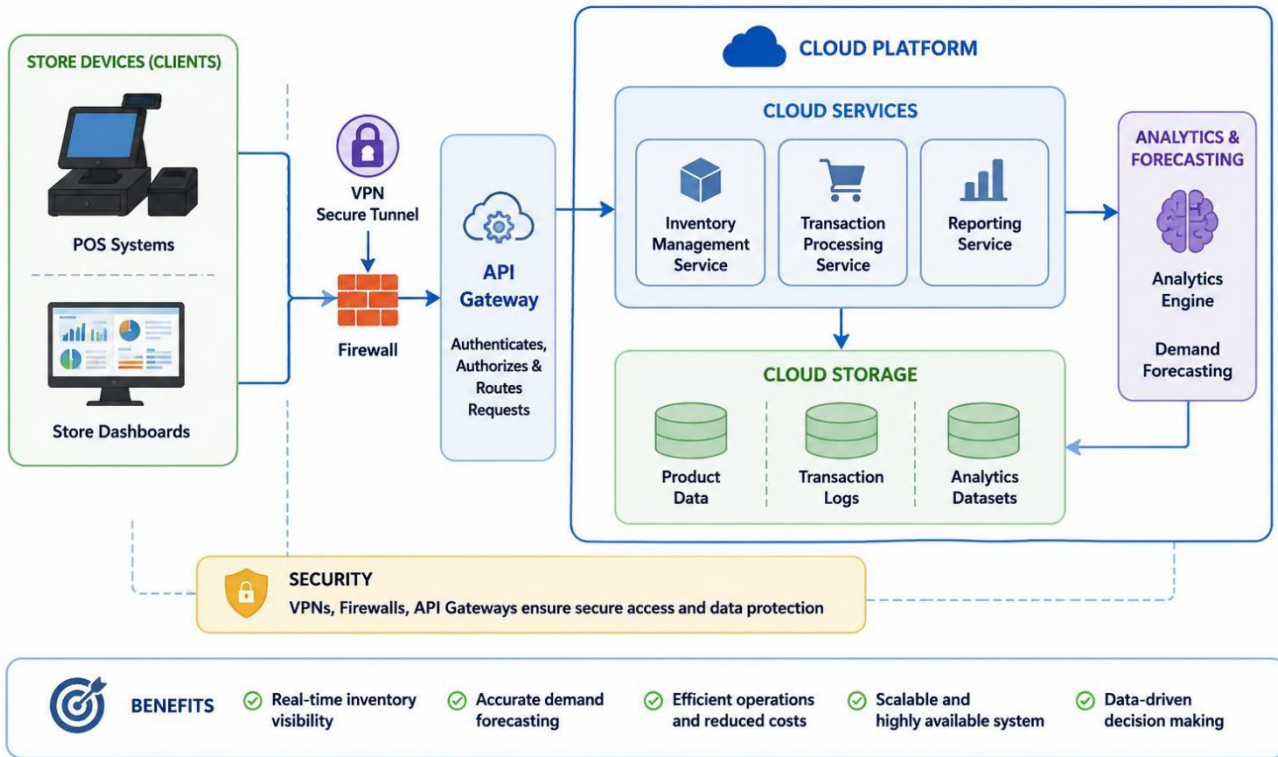
VPNs, firewalls, and API gateways provide secure access to cloud resources.

**6. Overall benefit:**

The cloud solution improves inventory management, analytics, forecasting, and operational efficiency.

## Case Study 1: Smart Retail Inventory Management System

### Cloud Architecture



## Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration. **Cloud-Based Document Collaboration System**

### 1. Cloud services used:

Distributed cloud storage, Web APIs, network infrastructure, and security services are used.

### 2. Role of cloud storage:

Documents are stored in distributed cloud storage with version control.

### 3. Role of Web APIs:

APIs synchronize documents and changes across multiple clients and devices.

### 4. Role of browsers:

Users access the platform through browsers for real-time document editing and sharing.

### 5. Role of network infrastructure:

It provides low latency and high availability for users worldwide.

### 6. Security mechanisms:

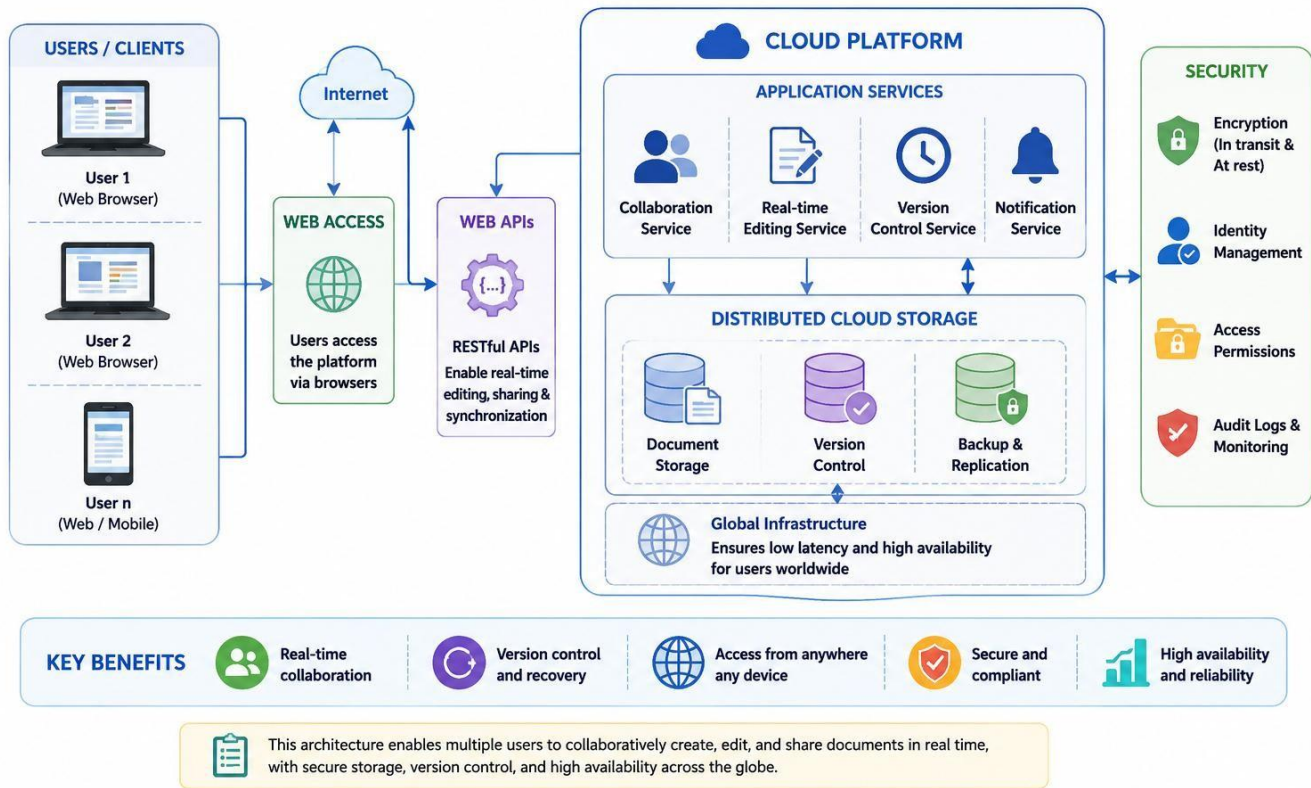
Encryption, identity management, and access permissions protect collaborative documents.

### 7. Overall benefit:

The system enables secure, real-time, and reliable document collaboration across devices.

## Case Study 2: Cloud-Based Document Collaboration System

Architecture Diagram



## Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time. **Online Food Ordering Platform**

### 1. Cloud services used:

Cloud platform services, RESTful Web APIs, cloud storage, virtual machines, and managed Kubernetes are used.

### 2. Role of RESTful APIs:

REST APIs connect web/mobile clients with backend cloud services.

### 3. Role of cloud storage:

It securely stores menu images, invoices, and customer data.

### 4. Role of virtual machines:

Virtual machines provide the computing infrastructure required to run backend services.

### 5. Role of Kubernetes:

Managed Kubernetes provides scalability and reliability for the application.

## 6. Security mechanisms:

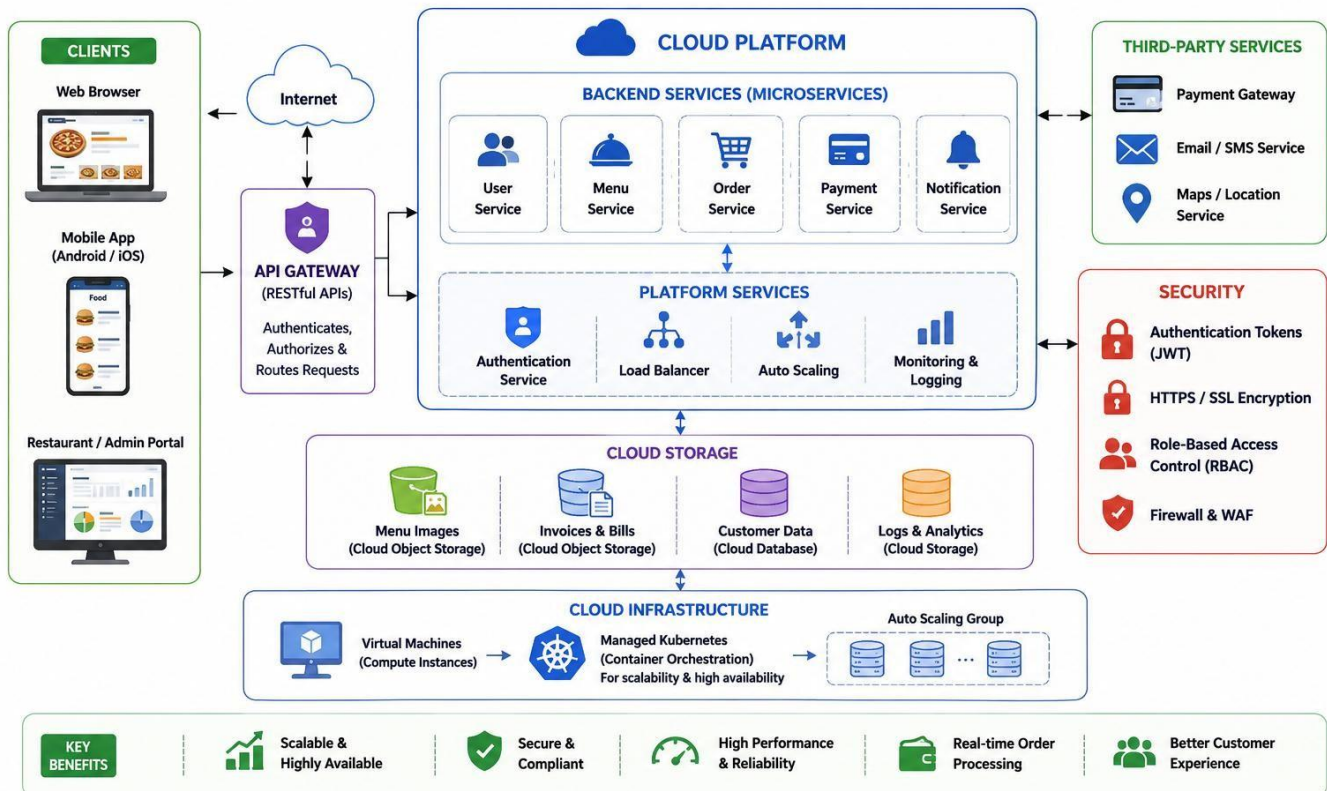
Authentication tokens, HTTPS, and role-based access control secure the platform.

## 7. Overall benefit:

The cloud architecture provides a scalable, reliable, secure, and real-time food ordering system.

# Case Study 3: Online Food Ordering Platform – Cloud Architecture

*A scalable, secure and reliable cloud-based solution*



## Case Study

## Main Cloud Focus

## Key Technologies

### 1. Smart Retail

Inventory & analytics APIs, cloud storage, analytics, forecasting

### 2. Document Collaboration

Real-time collaboration

Distributed storage, APIs, version control

### 3. Food Ordering Rubric

Scalable online service

REST APIs, storage, VMs, Kubernetes

Case Study

| Criteria              | Weightage | Level 4 - Exemplary                  | Level 3 - Proficient         | Level 2 - Developing     | Level 1 - Beginning        |
|-----------------------|-----------|--------------------------------------|------------------------------|--------------------------|----------------------------|
| Understanding of Case | 25%       | Clearly identifies all technical and | Identifies most requirements | Basic understanding only | Poor understanding of case |

|                         |     |                                                       |                                     |                     |                               |
|-------------------------|-----|-------------------------------------------------------|-------------------------------------|---------------------|-------------------------------|
|                         |     | business requirements                                 |                                     |                     |                               |
| Application of Concepts | 25% | Applies suitable cloud concepts and tools effectively | Good application with minor issues  | Partial application | Weak or incorrect application |
| Analysis                | 20% | Strong analysis with clear trade-off reasoning        | Reasonable analysis                 | Limited analysis    | Minimal analysis              |
| Solution Design         | 20% | Solution is feasible, practical, and well justified   | Reasonable solution with minor gaps | Basic solution      | Unclear solution              |
| Clarity                 | 10% | Wellstructured and easy to follow                     | Mostly clear                        | Somewhat unclear    | Poorly organized              |